Grade 9 Physical Sciences – Summary Notes

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The Physical Sciences paper is 1½ hours for 75 marks, covering Electricity, Forces, Magnetism and Electrostatics (Terms 3 and 4). Also learn the definitions on your class definition page.

Formula sheet

Quantity	Symbol	Unit	Formula
Charge	Q	coulomb (C)	Q = I × t
Current	I	ampere (A)	I = Q ÷ t
Potential difference	V	volt (V)	V = I × R
Resistance	R	ohm (Ω)	R = V ÷ I
Energy delivered by a cell	W (or E)	joule (J)	W = V × Q
Weight	w (or F_g)	newton (N)	w = m × g
Mass	m	kilogram (kg)	m = w ÷ g
Electrical energy used	—	kilowatt-hour (kWh)	kWh = power (kW) × time (h)
Cost of electricity	—	rand (R)	cost = kWh × price per kWh

- **Time must be in seconds** for Q = I × t (minutes × 60).
- **g = 10 N/kg** on Earth (some teachers use 9.8 N/kg — use what your teacher uses).
- Convert grams to kg (÷ 1 000) and watts to kW (÷ 1 000).
- Always write: formula → substitute → answer **with unit**.

Worked examples

1. A current of 2 A flows for 3 minutes. Q = I × t = 2 × 180 = **360 C**.
2. A 1.5 V cell moves 360 C. W = V × Q = 1.5 × 360 = **540 J**.
3. A 12 V battery pushes 0.5 A through a resistor. R = V ÷ I = 12 ÷ 0.5 = **24 Ω**.
4. A 60 kg learner: w = m × g = 60 × 10 = **600 N**.

5. A 2 000 W heater runs for 3 hours at R2.50/kWh: 2 kW × 3 h = 6 kWh; cost = 6 × 2.50 = **R15**.

Electricity 1: cells, generation and the grid

Cells and batteries

- A **cell** changes stored **chemical potential energy → electrical energy**. It has a positive (+) and negative (−) terminal and an electrolyte (paste or liquid).
- A **battery** is two or more cells connected together (in series = higher total voltage).
- A cell only delivers energy when it is in a **closed circuit**. When the chemicals are used up, the cell is "flat".
- **Energy conversion in a circuit:** chemical (cell) → electrical → light and heat (bulb), kinetic (motor), sound (buzzer).

Methods of generating electricity

Method	Energy conversion	Renewable?	Notes
Coal power station	Chemical → heat → kinetic (steam turns turbine) → electrical	No	Produces most of SA's electricity (Eskom); pollutes, CO_2
Nuclear	Nuclear → heat → kinetic → electrical	No	Koeberg near Cape Town; no CO_2 but radioactive waste
Hydro-electric	Gravitational potential (falling water) → kinetic → electrical	Yes	Needs dams and rivers
Wind	Kinetic (wind) → electrical	Yes	Only works when wind blows
Solar	Light (radiant) → electrical (solar panels)	Yes	Only works in daylight
Gas / diesel	Chemical → heat → kinetic → electrical	No	Used at peak times; expensive

In most power stations a **turbine** turns a **generator** (magnets and coils of wire) to produce electricity.

The national grid

The **national grid** is the network of power lines that carries electricity from power stations to users across the country.

1. **Power station** generates electricity.
2. **Step-up transformer** increases the voltage (up to ~400 000 V).
3. **High-voltage transmission lines** (pylons) carry it long distances — high voltage means **less energy is lost as heat**.
4. **Substation / step-down transformer** reduces the voltage.
5. **Distribution lines** take it to homes (230 V), schools and factories.

Cost of electricity

- Electricity is sold per **kilowatt-hour (kWh)** = one "unit" = using 1 000 W for 1 hour.
- **Cost = power (kW) × time (h) × price per kWh.**
- Appliances that **heat** (geysers, kettles, heaters, stoves) use the most electricity.
- **Save electricity:** switch off lights and appliances, use LED bulbs, put a geyser blanket on, use solar water heaters, boil only the water you need.

Electricity 2: circuits

Circuit components

Component	Function	Symbol (describe)
Cell	Source of energy	Long thin line (+) and short thick line (−)
Battery	Two or more cells	Several cell symbols joined
Switch	Opens or closes the circuit	Line with a gap and a lifted arm (open)
Light bulb	Converts electrical energy to light and heat	Circle with a cross
Resistor	Limits (opposes) the current	Rectangle
Variable resistor (rheostat)	Changes resistance and current	Rectangle with an arrow through it
Ammeter	Measures current; connected **in series**	Circle with A

Component	Function	Symbol (describe)
Voltmeter	Measures potential difference; connected **in parallel**	Circle with V
Conducting wire	Connects components	Straight line
Fuse	Melts and breaks the circuit if current is too high	Rectangle with a line through it

A circuit must be **closed** (complete) for current to flow.

Key terms

Term	Meaning	Unit	Measured with
Charge (Q)	Property of particles (electrons) that flow in a circuit	coulomb (C)	—
Current (I)	Rate of flow of charge (amount of charge per second)	ampere (A)	Ammeter (in series)
Potential difference (V)	Energy transferred per coulomb of charge between two points	volt (V)	Voltmeter (in parallel)
Resistance (R)	Opposition to the flow of current	ohm (Ω)	— (calculate R = V ÷ I)

Conventional current flows from **+ to −** outside the cell; electrons flow from − to +.

Series and parallel circuits

	Series circuit	Parallel circuit
Layout	Components one after another; **one path**	Components on separate branches; **more than one path**
Current	**Same everywhere**: $I = I_1 = I_2$	**Splits** between branches: $I = I_1 + I_2$
Potential difference	**Shared**: $V = V_1 + V_2$	**Same** across each branch: $V = V_1 = V_2$
Total resistance	**Adds up**: $R = R_1 + R_2$ (more resistors → more resistance)	**Decreases** when you add resistors in parallel
Adding more bulbs	Bulbs get **dimmer**	Bulbs stay **equally bright**

	Series circuit	Parallel circuit
One bulb breaks	**All** go out	Others stay **on**
Example	Old Christmas lights	Wiring in a house

For two resistors in parallel: $1/R = 1/R_1 + 1/R_2$ (total is always smaller than the smallest resistor).

Example (series): a 6 Ω and a 4 Ω resistor in series with a 5 V cell: R = 6 + 4 = 10 Ω; I = V ÷ R = 5 ÷ 10 = **0.5 A** (same through both); V across 6 Ω = 0.5 × 6 = 3 V, across 4 Ω = 2 V (3 + 2 = 5 V ✓).

Example (parallel): two branches carry 0.4 A and 0.6 A → total current from the cell = 0.4 + 0.6 = **1 A**.

Factors that affect the resistance of a conductor

Factor	Effect on resistance
Length	Longer wire → **more** resistance
Thickness (cross-sectional area)	Thicker wire → **less** resistance
Type of material	Copper and silver have low resistance; nichrome has high resistance
Temperature	Hotter wire → **more** resistance

Good **conductors** (metals) let current flow easily; **insulators** (plastic, rubber, wood, glass) do not.

Forces

A **force (F)** is a push or a pull on an object. The unit of force is the **newton (N)**. Force has size and direction, so it is drawn as an arrow.

4 effects of forces

A force can:

1. **Change the shape** of an object (squash, stretch, bend).
2. **Start or stop** an object moving.
3. **Change the speed** of an object (faster or slower).

4. **Change the direction** of a moving object.

Contact forces and field forces

- **Contact forces** act only when objects **touch**.
- **Field forces** (non-contact) act **at a distance** without touching.

Contact force	Description
Applied force	A push or pull by a person or object (e.g. pushing a trolley)
Friction	Opposes motion between two surfaces in contact; acts **opposite** to the direction of motion
Normal force	Push of a surface on an object resting on it; acts **perpendicular (90°)** to the surface, upwards on a table
Tension	Pull in a stretched rope, string or cable; acts **along** the rope, away from the object

Field force	Description
Weight (gravitational force)	Pull of the Earth on an object; always acts **downwards** towards the centre of the Earth
Electrostatic force	Attraction or repulsion between charged objects
Magnetic force	Attraction or repulsion between magnets, or a magnet and a magnetic material

Force diagrams and free body diagrams

- **Force diagram:** draw the object, with arrows showing all forces acting **on** it.
- **Free body diagram:** draw the object as a **dot**, with arrows starting at the dot.
- Rules: arrows point in the direction of the force; **longer arrow = bigger force**; label each arrow with its name or symbol (e.g. F_g / w, F_N, F_f, F_A, F_T) and size.

Example — a box being pushed to the right across the floor: weight (down), normal force (up, same length as weight), applied force (right), friction (left).

Example — a lamp hanging from a cable: weight (down), tension (up), equal in length.

Mass and weight

	Mass	Weight
Meaning	Amount of matter in an object	Gravitational force of the Earth on the object
Unit	kilogram (kg)	newton (N)
Measured with	Balance / scale	Spring balance (newton meter)
Changes on another planet or the Moon?	**No**, stays the same	**Yes**, depends on gravity (on the Moon it is about 1/6)
Type	Not a force	A field force

w = m × g, with g = 10 N/kg on Earth (gravitational field strength).

- 5 kg bag: w = 5 × 10 = **50 N**.
- An object weighs 300 N: m = w ÷ g = 300 ÷ 10 = **30 kg**.
- 500 g = 0.5 kg → w = 0.5 × 10 = **5 N**.

Magnetism

Magnetic force and poles

- Every magnet has a **north (N)** and a **south (S) pole**. The magnetic force is **strongest at the poles**.
- **Like poles repel** (N–N, S–S); **unlike poles attract** (N–S).
- Magnetic force is a **field force**: it acts without touching, and gets weaker with distance.
- Poles always come in pairs: if you break a magnet, each piece gets its own N and S pole.
- A freely hanging magnet (compass) turns so its N pole points to the Earth's geographic north.

Magnetic fields and field patterns

A **magnetic field** is the region around a magnet where it exerts a magnetic force. It can be shown with iron filings or a plotting compass.

Rules for drawing **magnetic field lines**:

- Lines go **from N to S** outside the magnet (arrows point away from N, towards S).
- Lines **never cross**.
- Lines are **closest together at the poles**, where the field is strongest.

Arrangement	Field pattern
Single bar magnet	Curved lines loop from N around to S on both sides
Unlike poles facing (N–S)	Lines run straight across the gap from N to S — **attraction**
Like poles facing (N–N)	Lines bend away from each other; a **neutral point** (no field) in the middle — **repulsion**

Magnetic substances and ferromagnetic materials

- **Magnetic materials** are attracted to a magnet. **Non-magnetic** materials are not (wood, plastic, glass, copper, aluminium, gold, silver).
- **Ferromagnetic materials** are strongly attracted and can be magnetised: **iron, nickel, cobalt** and their alloys (e.g. **steel**).
- **Soft iron** magnetises easily but loses its magnetism quickly (used in electromagnets). **Steel** is harder to magnetise but stays magnetised (used for permanent magnets).
- Magnetise by stroking with a magnet in one direction or with an electric current; demagnetise by heating, hammering or dropping.

Magnetic field of the Earth

- The Earth behaves like a giant bar magnet, because of the moving molten iron in its **outer core**.
- The Earth's **magnetic south pole is near the geographic North Pole** — that is why the N pole of a compass points north (unlike poles attract).
- The magnetic poles are not exactly at the geographic poles, and they move slowly over time.
- The magnetic field (magnetosphere) **protects Earth from the solar wind** (harmful charged particles from the Sun). Where these particles enter near the poles, they cause the **aurora** (northern and southern lights).
- Uses: navigation with a compass; animals such as birds and turtles use it for migration.

Electrostatics

Charges

- Atoms contain **protons (+)** in the nucleus and **electrons (−)** around it. A neutral object has equal numbers of protons and electrons.

- Charge is measured in **coulomb (C)**.
- Only **electrons move** from one object to another. Protons stay in the nucleus.

How objects become charged (transfer of electrons)

When two different materials are **rubbed together (friction)**, electrons move from one to the other:

Object	What happened	Charge
Object that **gains** electrons	More electrons than protons	**Negative (−)**
Object that **loses** electrons	Fewer electrons than protons	**Positive (+)**

Examples:

- Plastic ruler rubbed with a woollen cloth → ruler gains electrons (−), cloth loses electrons (+).
- Glass rod rubbed with silk → glass loses electrons (+), silk gains electrons (−).
- Balloon rubbed on hair → balloon (−), hair (+); the hair stands up and sticks to the balloon.

Electrostatic force

- **Like charges repel** (+ and +, − and −); **unlike charges attract** (+ and −).
- A charged object can also **attract a neutral object** (e.g. a charged ruler picks up small bits of paper).
- It is a **field force**, stronger when the objects are closer together and when the charges are bigger.

Conservation of charge

- **Charge cannot be created or destroyed; it is only transferred** from one object to another.
- The **total charge stays the same** before and after rubbing. If the ruler gains 5 electrons, the cloth loses exactly 5 electrons: total charge = 0 before and after.

Everyday examples

- Getting a small shock from a car door or a trampoline.
- Clothes sticking together after the tumble dryer.
- **Lightning** — a large discharge between charged clouds and the ground.

- Uses: photocopiers, spray painting, dust filters in chimneys.
- Dangers: sparks can ignite fuel (why petrol trucks are earthed with a chain or cable).